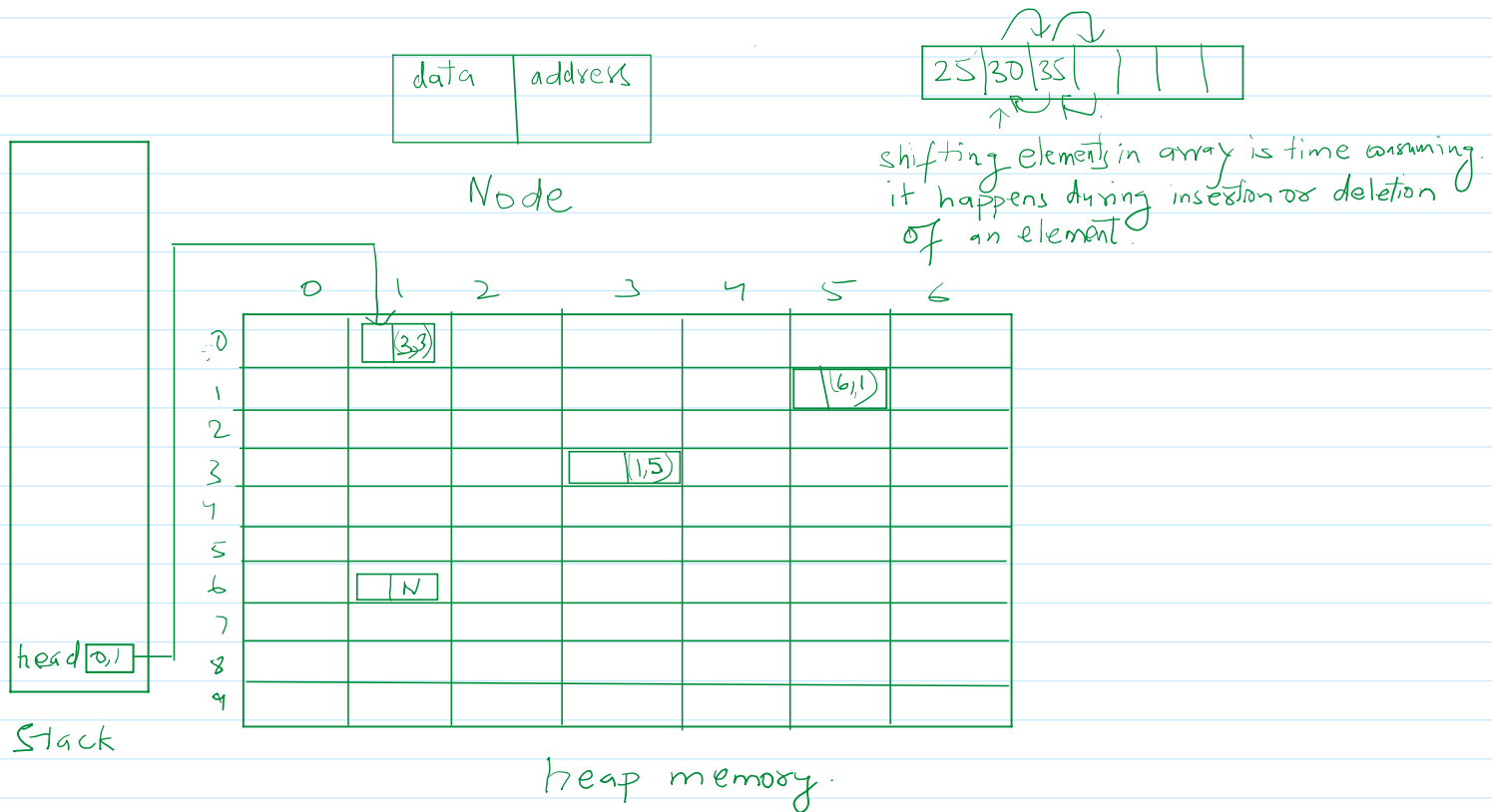


Linked List

02 September 2026 20:09

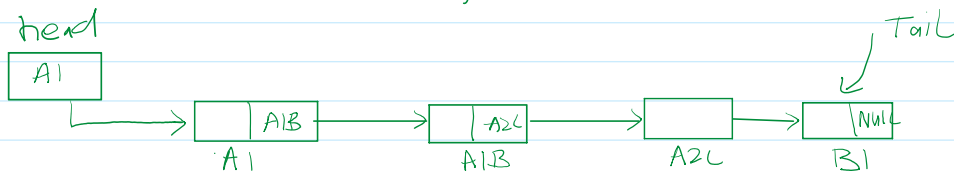
A linked list is a linear data structure that includes series of connected nodes. A linked list can be defined as a set of nodes that are randomly stored in memory. A node in the linked list contains 2 parts. That is the first is the data part and the 2nd is the address part. The last node of the list contains a pointer to NULL. After array linked list is the second most used data structure.

In a linked list, every link contains a connection to another link.



What is a Node?

A node is a self referential structure/class. It means, it can hold the reference of another node in it of its own type.



How to declare a Node?

```
struct Node {  
    int data;  
    struct Node * next;  
};
```

```

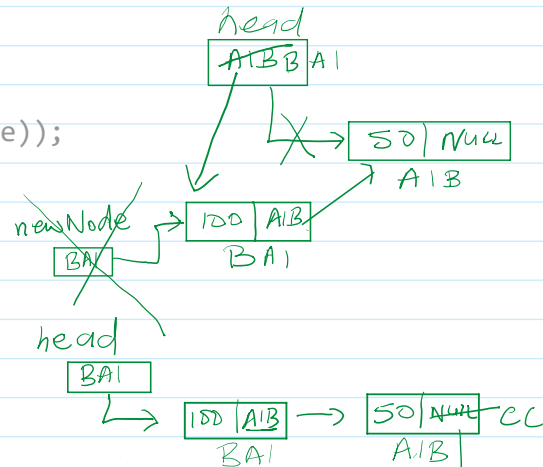
    struct Node *next;
};

```

```

void addNodeAtStart(Node **head, int value){
    → Node *newNode = (Node *)calloc(1, sizeof(Node));
    → newNode->data = value;
    → newNode->next = NULL;
    → if(*head == NULL){
        → *head = newNode;
        return;
    }
    newNode->next = *head;
    *head = newNode;
}

```



```

void addNodeAtEnd(struct Node **head, int value){
    struct Node *newNode = (struct Node *)calloc(1, sizeof(struct Node));
    newNode->data = value;
    newNode->next = NULL;
    → if(*head == NULL){
        *head = newNode;
        return;
    }
    → struct Node *t = *head;
    → for(; t->next; t = t->next);
    → t->next = newNode;
}

```

